

PULSE SA — EXECUTIVE SUMMARY

Auditable Alternative Credit Intelligence & Ground-Truth Infrastructure for South Africa's Informal Economy

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Memorandum

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Release)

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CLASSIFICATION:
Confidential / Commercial

THE CORE PROPOSITION: Pulse SA deploys a verified youth data-researcher network and AI Vision Neural OCR to capture 3-way audited financial and operational telemetry on unbanked township SMEs. We convert invisible cash-based trading activity into bank-grade alternative credit scorecards, de-risking closed-loop working capital loans and asset micro-insurance for South Africa's leading financial institutions.

1. The Market Failure: The R750 Billion Data Blindspot

South Africa's informal economy represents over **3.3 million micro-enterprises** generating an estimated **R750 Billion in annual turnover**. Despite representing over 40% of national economic livelihood and food security, more than 92% of these businesses are categorized as 'Thin File' or 'Unbankable' by traditional credit bureaus (TransUnion, Experian).

Institutional Requirement	Traditional Status Quo (Why Banks Fail)	Pulse SA Audited Solution
Cash Flow Verification	Demands 6 months stamped bank statements & audited financial accounts (non-existent in cash economy).	Inverted COGS Reconstruction: AI Vision OCR scans formal wholesale till slips (Metro, Boxer) + 25-35% retail markup modeling.
Business Existence & Fraud	Blind desktop algorithms; high risk of ghost borrowers and fabricated addresses.	Physical Hardware Audit: GPS coordinate clustering, timestamped on-site photo capture, and repeat agent visits.
Repayment Discipline	Zero formal credit history; instant rejection by automated Loan Origination Systems (LOS).	Alternative Peer Track Record: 3+ years verified stokvel contributions, burial society records, and supplier trade terms.
Loan Disbursement Risk	Cash loans deposited into personal accounts are diverted to household emergencies, leading to default.	Closed-Loop Stock Vouchers: Digital SMS vouchers redeemable exclusively for inventory at partner wholesalers.

2. Commercial Product Verticals

Pulse SA monetizes its ground-truth data infrastructure through three high-margin enterprise data products:

Vertical A: SME Alternative Credit Scoring (For Commercial Banks & Fintechs)

Powers uncollateralized working capital facilities for lenders such as Capitec Business, Standard Bank, TymeBank, and Retail Capital. We generate a dynamic **Pulse Credit Score (0-100)**, assign risk tiers (Tier A to C), and calculate exact credit limits (R3,000 to R25,000) based on verified monthly turnover, supplier match rates, and operating longevity.

Vertical B: Risk & Asset Micro-Underwriting (For Insurers & Underwriters)

Enables short-term insurers (Hollard, Santam, Old Mutual) to write low-premium parametric micro-policies (R60 to R180/month) for township containers and spazas. We audit premises construction materials (brick vs owned container vs shack), fire hazards, flood vulnerability, and load-shedding meat spoilage exposure.

Vertical C: ESD & DFI Impact Telemetry (For Corporate Funds, DFIs & Donors)

Provides auditable monitoring for Enterprise Supplier Development (ESD) funds, SEFA, NEF, and provincial economic development agencies. Delivers transparent B-BBEE scorecard compliance, tracks youth stipend disbursements, and measures township formalization trajectories across 4,488 municipal wards.

3. Forensic Cash Audit Methodology: Input-to-Output Triangulation

Under **International Standard on Auditing (ISA) 500 & ISA 240**, uncorroborated cash sales represent high inherent risk. Pulse SA overcomes this through an empirical 3-Factor Triangulation Model:

Audit Factor	Mechanism & Evidence Capture	Mathematical Verification Formula
1. Upstream Wholesale Invoices (COGS)	AI Vision Neural OCR scans formal printed tax invoices from Metro, Boxer, and Abattoirs. Extracts VAT numbers, dates, and itemized SKUs.	Reconstructed Revenue = Audited Wholesale Invoices × (1 + Standard Township Markup %)
2. Longitudinal Foot-Traffic Audit	Independent spot-checks by multiple agents across morning, midday, and evening trading windows confirm active customer volume.	Cash Velocity = Audited Customers/Hr × Trading Hours × Avg Kasi Basket Size (R35)
3. Route Distributor Drop-Sheets	Photo verification of daily bread (Sasko/Albany) and beverage (Coca-Cola) bakkie delivery carbon notes.	Daily Inventory Turnover = Units Delivered Daily × Unit Retail Selling Price

4. Technology Stack & Integration Architecture

Pulse SA is built as an end-to-end distributed data platform designed specifically for emerging market connectivity constraints:

- **Field Agent PWA (Progressive Web App):** Zero-data-cost architecture with full offline IndexedDB caching. Features dynamic audio/visual prompts, hardware GPS lock, and camera compression.
- **AI Vision Neural OCR:** Automated slip recognition engine that parses skewed, crumpled till slips, verifies VAT registration authenticity, and normalizes SKU descriptions.
- **Algorithmic DQS Engine (0-100):** Tests mandatory field completeness, GPS spatial clustering, price plausibility bounds, and internal arithmetic consistency before pipeline ingestion.
- **Municipal Demarcation Board (MDB 2026) Layer:** Full geospatial integration across all 4,488 official South African election wards for hyper-local credit risk mapping.
- **Institutional API & Webhook Dispatch:** Secure REST endpoints (`/v1/traders/{id}/credit-profile``) and automated SFTP batch delivery for core banking Loan Origination Systems.

5. Commercial Pilot Deployment & Unit Economics

Metric / Parameter	Pilot Phase (Soweto Ward 15)	Metro Scale (Gauteng 100 Wards)
Active Field Youth Researchers	25 Verified Agents	500 Verified Agents
Trader Dossiers Profiled / Month	250 Fully Audited SMEs	5,000 Fully Audited SMEs
Agent Acquisition & Reward Cost	R15 - R35 per completed profile	R12 - R25 per completed profile
Institutional Credit Dossier Fee	R150 - R250 per scored profile	R120 - R200 per scored profile (API SaaS)
Gross Margin on Data Asset	78.5% Gross Margin	84.2% Gross Margin

6. Regulatory Governance, POPIA & Institutional Trust

Pulse SA operates under strict adherence to the **Protection of Personal Information Act (Act 4 of 2013)** and the **Electronic Communications and Transactions Act (Act 25 of 2002)**:

- **Informed Trader Consent:** Verbal and written consent protocols explaining commercial data utilization prior to profile capture.
- **Visual Identity Verification:** RSA ID documents are inspected visually for existence and fraud verification; plain-text national ID numbers are never sold or exposed in public datasets.
- **Audit Trail Integrity:** Immutable cryptographic hashing of all survey submissions, GPS coordinates, and receipt timestamps to ensure institutional evidential admissibility.

INSTITUTIONAL ENGAGEMENT & PILOT INQUIRIES:

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